Fine Mapping and Dynamic Analysis of Grassland–Livestock Balance in Alpine Grassland From 2019 to 2022 Based on RAV-Satellite Spectral Fusion and Statistical Data Spatialization

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***Abstract*— Assessing the state of the grassland–livestock balance is of great practical significance for continuously optimizing grazing management and ensuring the sustainable use of alpine grasslands. However, obtaining fine, dynamic, and continuous information of the grassland–livestock balance in alpine grassland is greatly hindered due to limitations in precision, timeliness, and spatial scale of two key indicators, theoretical livestock carrying capacity (TLCC) and actual livestock carrying capacity (ALCC). On the one hand, this study constructed grassland aboveground biomass (AGB) estimation models using the multispectral remote aerial vehicle (RAV) data and Sentinel-2 data in the northeast of Tibetan Plateau based on the nonnegative matrix factorization (NMF)-based differentiated fusion method. The annual high-precision TLCC spatial distribution dataset was obtained through simulation from 2019 to 2023. On the other hand, this study constructed livestock density estimation models based on the livestock statistical data of the smallest administrative unit and various gridded environmental and socioeconomic data from 2019 to 2022; it thus achieved the fine spatialization of the regional-scale ALCC. The spatiotemporal dynamics of TLCC and ALCC sequences were comprehensively analyzed, and the grassland–livestock balance status and its driving factors from 2019 to 2022 were further evaluated in the study area. We concluded that: 1) the R^2 of the optimal RAV-based and Sentinel-2-based grassland AGB models reached 0.88 and 0.77, with corresponding root mean square error (RMSE) of 552.39 and 692.41 kg/ha, respectively; 2) differentiated fusion with RAV spectral data significantly enhanced the estimation accuracy of Sentinel-2 data for regional-scale grassland AGB (R^2 improved from 0.77 to 0.80); 3) the RF-based livestock density model had the best spatialization effect on ALCC ($R^2 = 0.80$ and RMSE = 77.34 sheep/km^2); 4) from 2019 to 2022, the overall status of the grassland–livestock balance was good, with over 65% of the grassland area consistently nonoverloaded or balanced, while severely overloaded grassland area remained below 16.14%; large interannual fluctuations were observed in nonoverloaded and severely overloaded grasslands; the overloaded areas mainly concentrated in the core pasture and the periphery of high population density areas; and 5) livestock density, temperature, and distance to river were the main driving factors of livestock–grassland balance dynamics. Innovatively, an annual 10-m spatial resolution grassland–livestock balance dataset was developed.**

***Index Terms*— Actual livestock carrying capacity (ALCC) spatialization, data fusion, spatiotemporal dynamics, theoretical livestock carrying capacity (TLCC) modeling, variable diversification.**

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I. Introduction

ALPINE grassland dominates the Tibetan Plateau, covering approximately 70% of its total area [1]. It is crucial for local animal husbandry and herders' livelihoods [2], [3]. Accurately assessing the grassland–livestock balance is beneficial for optimizing the grazing system, managing grassland resources efficiently, and ensuring the sustainability of grassland animal husbandry on the Tibetan Plateau. However, due to data and monitoring technology constraints, accurately obtaining fine, dynamic, and continuous information on grassland–livestock balance in alpine environments remains extremely challenging.

The prerequisite and basis for assessing the grassland–livestock balance is to obtain accurate theoretical livestock-carrying capacity (TLCC) and actual livestock-carrying capacity (ALCC) data. The TLCC represents the maximum livestock numbers that ensure normal livestock growth, reproduction, and production while maintaining sustainable grassland production [4]. It is usually based on the aboveground biomass (AGB) of the grassland, which is considered the most direct and effective method [5]. Therefore, accurate grassland AGB estimation is a critical step for obtaining TLCC. Although

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the feasibility of using remote sensing spectral variables and machine learning to estimate the regional-scale grassland AGB has been demonstrated, most studies use a single type of modeling variables, considering the spectral but not structural information of the vegetation. However, vegetation structure and heterogeneity are closely related to texture features. Texture measures the variability of pixel values between neighboring pixels; it describes the spatial relationships and variations between pixels and is more complex than spectral information [6], [7]. Therefore, the synergistic use of spectral and texture features facilitates the construction of more accurate AGB estimation models than spectral features alone [8]. For example, Grüner et al. [6] constructed an AGB estimation model for the mixed farmland of Leguminosae and Gramineae using spectral and texture features from remote aerial vehicle (RAV). Their result showed that the addition of texture features greatly improved the prediction accuracy of cropland AGB (0.62 versus 0.86). Kelsey and Neff [9] studied the San Juan National Forest Park and estimated the forest AGB using vegetation indices and texture features of the Landsat TM. Their results indicated that including texture features improved the model's prediction accuracy by more than 50%. However, texture features are mainly applied for the biomass estimation of forests and croplands but rarely for grassland AGB estimation studies. Their estimation potential remains to be further explored.

Most regional-scale grassland AGB estimations are based on medium-resolution or low-resolution remote sensing images, which are susceptible to the influence of mixed pixels and have poor spatial matching with the measured data. Therefore, they make obtaining fine grassland AGB inversion results challenging, thus affecting the accurate estimation of TLCC. In comparison, high-resolution remote sensing data have obvious advantages in estimating grassland AGB [10], [11]. For example, the Sentinel-2 multispectral instrument (MSI) data are freely accessible and provide high spatiotemporal resolution. It is particularly noteworthy because it is the only multispectral satellite that includes three bands within the red edge range [12]. The red edge band is sensitive to vegetation growth status, which is more advantageous for accurately monitoring vegetation AGB [13], [14], [15]. Additionally, the rapid advancement of RAV technology has made multispectral RAV become another favorable tool to obtain information on grassland vegetation [16]. Compared with satellite methods, RAV is less affected by atmospheric interference, has a surface resolution at the centimeter level or even millimeter level, and enables the real-time transmission and acquisition of images [17], [18]. Grassland AGB models constructed by multispectral RAV data usually achieve higher accuracy than grassland AGB models constructed by satellite data [19], [20]; however, limited by battery endurance, RAV is only suitable for small scales (e.g., sample plots or home pastures) and cannot meet large-scale and long time-series grassland AGB monitoring. Satellite and RAV remote sensing have advantages and disadvantages, and the synergistic application of the two may effectively solve the above problems.

The fusion of RAV and satellite can be divided into three aspects: image fusion, model fusion, and spectral index fusion [21], [22], [23], [24], [25], [26], [27]. Spectral index fusion is the fusion of spectral indices from different sensors to produce new spectral indices; it has become a mainstream remote sensing data fusion method and has been widely used due to its low computation volume, fast processing, and good fusion effect [28], [29]. In recent years, the nonnegative matrix factorization (NMF) algorithm was applied to study the fusion of remote sensing spectral indices. It exhibits simple implementation, intuitive decomposition form, and good interpretability. Additionally, different samples' heterogeneous relationships can be revealed by the NMF algorithm, reasonably select the dimensions of features, and obtain the detail information contained in the original data to get the fused spectra with high-detail retention and good performance [30], [31], [32], [33]. Chen et al. [34] used the NMF algorithm to decompose and reconstruct the spectral variables of the ground-based hyperspectral and Sentinel-2 MSI. Their result indicated that this algorithm could effectively improve the correlation between Sentinel-2 spectral variables and soil salinity, and the coefficient of determination (R^2) of the fused model was improved by 0.09. Guo et al. [35] focused on the grassland AGB during the wilting stage and observed that the NMF-based differentiated fusion method could improve satellite-based estimation accuracy by 11.1%. These studies demonstrate the significant benefits of the NMF algorithm in fusing ground-based hyperspectral data and satellite data. However, whether this algorithm is equally applicable to the fusion between multispectral RAV and satellite data remains inconclusive and requires further verification.

ALCC is another important indicator in assessing grassland–livestock balance. It is defined as the actual number of grazing livestock in a specific area. ALCC data mainly derive from statistical animal husbandry data and field or household surveys conducted by scientific researchers. However, manual surveys are usually time-consuming and involve a huge amount of work, making monitoring ALCC dynamics at the regional scale impractical, and greatly limited by survey costs, timeliness, and spatiotemporal scales. Existing livestock statistical data in China mostly use administrative divisions as statistical units. In the past, relevant studies and applications of grassland–livestock balance usually used the number of year-end livestock in the statistical yearbooks as the real ALCC value. Therefore, using statistical data cannot analyze the actual distribution and changes of ALCC at smaller scales (e.g., pastureland or satellite image pixels), resulting in the inability to match with pixel-scale TLCC data and greatly restricting the precision and accuracy of the dynamic analysis of the grassland–livestock balance [36], [37], [38].

Fine-scale analyses of ALCC can be effectively achieved by decomposing livestock numbers within an administrative area into spatial grids using statistical data and spatialization techniques [39], [40]. Neumann et al. [41] used empirical and expert scoring methods to downscale and decompose European livestock statistical data into 1×1 km spatial grids to obtain spatial distribution maps of ALCC in various European countries. However, the expert scoring method relies strongly on individual knowledge and experience and is susceptible to subjective factors, leading to a certain lack of objectivity and reproducibility of the results. Generally, the livestock's distribution and number are mainly influenced

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by environmental and socioeconomic factors, and the spatial decomposition of livestock statistical data using gridded environmental and socioeconomic data has a stronger scientific basis and interpretability [42]. Robinson et al. [43] constructed a spatial dataset of global ALCC based on geographic data, such as land cover and population density, combined with the multilayer linear regression algorithm and verified that the R^2 of models in different regions ranged from 0.18 to 0.66. Li et al. [44] selected ten environmental factors and three socioeconomic factors and used machine learning to implement the spatialization of ALCC in western China, and the R^2 of the optimal model reached 0.75. However, most existing ALCC datasets are based on global or national scales, and their accuracy at small spatial scales cannot be guaranteed [45]. Additionally, the existing studies generally suffer from poor timeliness of statistical data, discontinuity in the time series of spatialization results, and coarse spatial resolution. Therefore, achieving the fine spatialization of ALCC at the regional scale based on the annual livestock statistical data of the smallest administrative unit and combining it with various environmental and socioeconomic factors is urgently required to accurately analyze grassland–livestock balance.

This study focuses on the alpine grassland in Menyuan County, eastern Tibetan Plateau, and explores the high-precision estimation method of grassland TLCC and ALCC at the county scale. It aims to create a high spatial resolution and continuous time-series datasets of TLCC and ALCC to accurately evaluate the dynamics of grassland–livestock balance. The specific objectives are: 1) exploring the effectiveness of texture features in grassland AGB estimation; 2) creating a high-precision grassland AGB estimation model by fusing RAV and satellite data; 3) implementing fine spatialization of ALCC at the regional scale using livestock statistical data and environmental and socioeconomic factors; and 4) analyzing the spatiotemporal change characteristics of TLCC and ALCC at the pixel level and quantitatively discussing the grassland–livestock balance status of alpine grassland in Mengyuan County in 2019–2022.

II. Data and Methods

A. Study Area

This study was executed in Menyuan County, Haibei Tibetan Autonomous Prefecture, Qinghai Province, China (100°55′E–102°38′E, 37°03′N–37°52′N) [Fig. 1(a)]. The administrative area of this county is about 6902 km^2, the elevation ranges between 2388 and 5235 m, and the climate is typical highland continental. The annual average temperature of Menyuan County is between −12 °C and 3 °C, annual precipitation exceeding 500 mm in most areas [46]. The natural grassland area in Menyuan County accounts for 63% of the total area, with alpine grassland accounting for over 95% [Fig. 1(b)]. Due to the rich grassland resources, Menyuan County vigorously develops animal husbandry. Their feeding method is mainly grazing.

B. Data Acquisition and Preprocessing

1) Grassland Field Data: Field observations occurred during the grassland growing seasons (July–September) from 2021 to 2023, and 186 sample plots were surveyed and collected during these three years. We employed a survey strategy that combined fixed and random sample plots. The setting of sample plots (e.g., size, number, and spacing, etc.) strictly adhered to the "technical specification for investigation and assessment of national ecological status: field observation of grassland ecosystem (HJ 1168–2021)" [47]. The spatial distribution of the sample plots is presented in Fig. 1(c). The two fixed sample areas are located in the National Field Scientific Observation Station for the Alpine Grassland Ecosystem (abbreviation: Haibei Station) in Haibei, Qinghai, and each sample area contains 18 fixed sample plots [Fig. 1(d)]. Random sample plots were set up using a stratified semi-random sampling method in alpine grassland communities [47], [48]. The semi-random in our context means that plots have been selected based on the constraint that surrounding vegetation must be homogeneous to reduce spatial scale effects [48]. To ensure that the random samples adequately represented different grassland conditions in the study area, we selected sample plots from different regions, altitudes (2586–3841 m), slopes, aspects, and utilization intensities while ensuring accessibility and flatness of the grasslands. After collation and validation, 164 valid samples were used to build the grassland AGB model. More detailed information about the descriptive statistics of samples can be found in the Supplementary Material. During field observations, using GPS to note each sample plot's longitude, latitude, and elevation. According to the size, community homogeneity, and species richness, 3–5 quadrats (0.5 × 0.5 m) in each sample plot were set up as appropriate. The canopy height, grassland coverage, chlorophyll content, and other indicators were measured. The above-ground part of the grassland was mowed flush with the ground, dried at 65 °C for 48 h, and the AGB dry weight was determined. Finally, the AGB dry weight average was calculated from quadrats within each plot to represent the grassland AGB value for the entire sample plot.

2) RAV Multispectral Data: Near-surface multispectral data were acquired using the multispectral version of

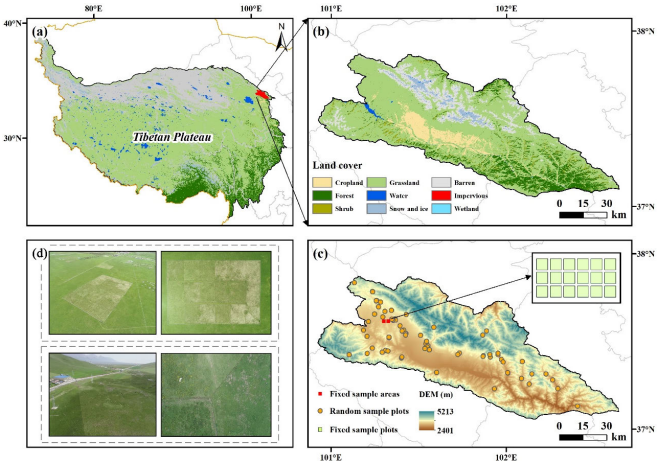


Fig. 1. Overview of the study area. (a) Geographic location, (b) land cover types, (c) DEM and distribution of sample plots, and (d) landscape photographs of sample plots taken by RAV (top to bottom: fixed sample plot and random sample plot).

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Phantom 4 RAV made by DJI (see Table II in the Supplementary Material for details). The RAV image-capture time was chosen as 10:00–15:00 when there is sufficient sunshine, and dry, windless, sunny, and less cloudy environments were preferred. The DJI GS Pro ground-station application was used to perform full-coverage flight imaging of each sample plot and perform radiative correction to the RAV whiteboard. The altitude was 50 m, the flight speed was 5 m/s, and the heading and lateral overlaps were controlled at 60% or more. During data processing, Pix4Dmapper software was used for lens distortion correction, radiative calibration, and image mosaic of RAV images. Next, ENVI 5.3 software was used for geometric correction and georeferencing, and the high-resolution RAV orthophoto of each sample plot was obtained. The reflectance value of each band corresponding to each sample plot's geographic location was further extracted, and the vegetation indices sensitive to the grassland AGB were calculated (see Table III in the Supplementary Material for details).

Each band's texture features were calculated based on RAV orthophotos. All texture features were normalized, helping reduce noise due to soil reflections and changes in the sun position, thus improving the estimation effect of the biomass [6]. Eight indicators from the gray level co-occurrence matrix were selected as texture features (see Table 4 in the Supplementary Material for details). These indicators have demonstrated excellent results in the existing vegetation AGB estimation studies [49]. For example, the eight texture features calculated for the blue band are: blue (energy), blue (entropy), blue (correlation), blue (inverse difference moment), blue (inertia), blue (cluster shade), blue (cluster prominence), and blue (Haralick correlation), respectively. When computing the texture features, choosing the right window size is critical for texture analysis; a too-small window may exaggerate information, while a too-large window may obscure important changes between pixels [50]. We set up five window sizes (3×3, 5×5, 7×7, 9×9, and 11×11) based on previous studies. To determine the optimal window size, we constructed grassland AGB models using each window size and compared their performance. The window size which resulted in the highest accuracy was selected as the optimal window for subsequent analysis. Texture features were computed in Python and completed by the Haralick Texture Extraction processing tool in the Orfeo Toolbox.

3) Sentinel-2 Satellite Data: Sentinel-2 MSI data were obtained from the ESA official website and Google Earth Engine. The product contains 13 spectral bands, with a minimum spatial resolution of 10 m. Sentinel-2 MSI includes two satellites, 2A and 2B; when both satellites operate simultaneously, a complete imaging task can be accomplished in five days, and the revisit cycle can be shortened to three days in some midlatitude regions. Images were selected based on the closest imaging time to the field observation time and low-cloud cover (<10%). The Sen2cor tool provided by ESA was used for atmospheric correction, radiative calibration, and cloud removal and then resampling all bands to 10 m. The reflectance values of each band corresponding to the geographic location of each sample plot were further extracted, and the vegetation indices sensitive to the grassland AGB were calculated (see Table III in the Supplementary Material for details). Additionally, eight texture features were calculated for each band at 10- and 20-m spatial resolutions based on Sentinel-2 images and computed through the same process as the RAV image texture features.

4) Environmental and Socioeconomic Data: The ALCC statistical data were obtained from the verification summary report of livestock carrying capacity in Menyuan County from 2019 to 2022, compiled by the Forestry and Grassland Administration of Menyuan County. These data were based on the village-scale number of year-end livestock, which was converted to standard sheep units according to Chinese Forestry and Grassland Industry Standards [51]; 436 statistical ALCC data within the study area's villages were obtained from 2019 to 2022, and the village-scale livestock density (i.e., ALCC per unit area) was calculated in combination with each village's administrative area.

The basic geodatabase used in the ALCC spatialization process included: land cover dataset (CLCD), nature reserve data, DEM data, river system dataset, monthly accumulated precipitation dataset, monthly average temperature dataset, population spatial distribution dataset (LandScan), nighttime lights dataset (DMSP-OLS), and normalized difference vegetation index (NDVI) dataset. More detailed information can be found in the Supplementary Material. Ten environmental factors were calculated and extracted from the above data, including grassland coverage, woodland coverage, cropland coverage, bare land coverage, elevation, slope, distance to river, annual cumulative precipitation, annual mean temperature, and annual maximum NDVI; and two socioeconomic factors were also determined: population density and nighttime light intensity. The above environmental and socioeconomic factors have different degrees of influence on the livestock's spatial distribution and are representative for simulating its spatial distribution characteristics [44], [52]. For subsequent calculations and analyses, all geographic data coordinate systems were processed uniformly as WGS1984 and resampled to a uniform spatial resolution of 10 m, which is consistent with the resolution of the Sentinel-2 MSI data.

Suitability masks are also a crucial issue to be considered in ALCC spatialization [53]. Considering the study area's characteristics, we chose a relatively conservative suitability masking strategy that only excluded water bodies, snow and ice, wetlands, core areas in the nature reserve where grazing is strictly prohibited, and high-density areas exceeding 1000 people/km^2 [44]. The environmental and socioeconomic data were masked according to the above strategy, after which each factor's mean values at the village scale were obtained by zonal statistics for the subsequent construction of the livestock density model. These calculations and processing were realized using ArcMap 10.6 software.

C. Construction and Evaluation of the Grassland AGB Model

To accurately calculate TLCC, a reliable estimation of grassland AGB is critical [36]. Constructing the grassland AGB model is an effective way to estimate grassland AGB at the

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regional scale, which is more efficient and comprehensive than the traditional method. For AGB prediction, a single model was established using all field data for different years instead of building a separate model for each year. We determined optimal AGB models at the plot and county scales through various variable selection methods and machine learning algorithms. The differentiated fused model was then constructed by fusing spectral variables from RAV and Sentinel-2 data. Both variable selection and spectral fusion were necessary processes for determining the optimal model.

1) Variable Selection: A total of 226 basic variables were involved to construct the grassland AGB models subsequently, including the respective original spectral bands of RAV and Sentinel-2 and the vegetation indices and texture features calculated based on those bands. Considering the possible collinearity between the variables and the model's complexity, the data redundancy was reduced by variable selection in constructing the grassland AGB estimation model to improve its accuracy and computational efficiency. Four methods were used to select all basic variables: genetic algorithms (GAs), least absolute shrinkage and selection operator (LASSO), successive projections algorithm (SPA), and stepwise regression (STEP). The model effectiveness of the selected variables from each method was then compared. We conducted all variable selection processes using MATLAB (R2021a). GA and SPA were realized by self-written programs, and LASSO and STEP were executed using the "lasso" and "stepwisefit" functions in MATLAB, respectively.

2) Modeling and Testing of the Machine Learning Model: Three machine-learning algorithms (RF, SVM, and ANN) were used to construct grassland AGB models based on RAV, Sentinel-2, and the synergy of the two, respectively. More detailed information about the introduction and parameter determination of the above three machine learning models can be found in the Supplementary Material. These machine learning algorithms were built using the scikit-learn library in Python.

The training and testing sets were divided using the ten-fold cross-validation method. The model performance was then evaluated by averaging the results from ten iterations. The evaluation metrics included coefficient of determination (R^2), root-mean-square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE) [(1)–(4)]. Each model was run 200 times by determining the random seeds, and the final model took the best result out of the 200 iterations

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - y_i)^2}{n}} \tag{2}$$

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n}|x_i - y_i|}{n} \tag{3}$$

$$\mathrm{MAPE} = \frac{\sum_{i=1}^{n}\left(\frac{|x_i - y_i|}{y_i}\right) \times 100\%}{n} \tag{4}$$

where x_i and y_i are the estimated and measured AGB values, respectively, $\bar{y}$ is the mean of the measured AGB values, and n is the sample size.

3) Differentiated Fusion of Spectral Variables Between RAV and Sentinel-2: By comparing and evaluating the models, the optimal RAV model and optimal Sentinel-2 model constructed based on selected variables were determined, respectively. It should be noted that since the bands of RAV and Sentinel-2 are not completely consistent, only spectral variables common to both can be fused. The original RAV spectral variables and the corresponding spectral variables used in the optimal Sentinel-2 model were differentially fused using NMF, and the RAV and Sentinel-2 spectral variables that satisfied the fusion condition were recorded as RAVSVs and MSISVs, respectively. As a new factor extraction method, NMF can decompose the original matrix into two nonnegative matrices: the feature matrix, which contains the rich spectral information, and the abundance matrix, which represents the contribution degree of different spectral features to the sample. The product of the two matrices approximates the original matrix [54]. In this study, RAVSVs and MSISVs could be viewed as an $m \times n$ original matrix V, where m is the number of features and n is the number of samples. A factorization factor number r (r is a positive integer) was predefined to determine the data size and level of approximation to the original matrix; it was essential for the decomposition effect of the matrix [35]. NMF could decompose the matrix V into W ($m \times r$) and H ($r \times n$), where W was the feature matrix and H was the abundance matrix. The factorization factor number was determined by the peak signal-to-noise ratio (PSNR); after testing, the optimal decomposition factor number was 3 [(5) and (6)].

Considering that the RAV image was less disturbed by the atmosphere and had a higher spatiotemporal resolution than the sentinel image, which had a larger swath width and stronger spatial representativeness. Therefore, W_{RAVSV} was used as the feature matrix and H_{MSISV} as the abundance matrix to combine the advantages of the two matrices. Optimal W_{RAVSV} and H_{MSISV} were obtained by alternately decomposing RAVSVs and MSISVs with the multiplicative update algorithm. Finally, multiplying these yielded the differentiated fused spectral variables (DFSVs). The relevant calculations were realized using the NMF function of scikit-learn in Python

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_i^2}{\mathrm{mse}_i}\right) \tag{5}$$

$$\mathrm{mse} = \frac{1}{n}\sum_{i=1}^{N}(V - W_{m\times r} \times H_{r\times n})^2 \tag{6}$$

where m is the number of features, n is the number of samples, r is the number of factorization factors, i is the number of main factors, and MAX_i is the maximum spectral input value of i.

4) Evaluation of the Effect and Feasibility of Differentiated Fusion: Twelve models were constructed based on Sentinel-2 original variables, different variable selection methods, and machine learning algorithms to determine the optimal AGB model. The MSISVs were replaced by DFSVs after the differentiated fusion, and the differentiated fused model was built

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by combining the model's remaining variables. The estimation performance of the optimal RAV model, the optimal Sentinel-2 model, and the differentiated fused model for grassland AGB were further compared to assess the fusion effect. Additionally, we analyzed the consistency between the RAVSVs and the MSISVs while comparing the correlation change between each spectral variable and grassland AGB before and after fusion to verify the fusion feasibility of RAV and Sentinel-2 spectral variables.

D. Construction and Evaluation of Livestock Density Model

Twelve environmental and socioeconomic factors were involved in constructing the livestock density model, all of which were the average values at the village level. The livestock density model was also constructed using RF, SVM, and ANN. Using the ten-fold cross-validation method to evaluate the models' performance. The evaluation indices were R^2 and RMSE.

Because the livestock density model was constructed using the factors' average values and livestock densities at the village scale, considering how factor values are distributed differently at the village and grid scales, directly applying it to the grid scale might result in errors. Therefore, the simulated distribution of livestock density required adjustment according to the actual number of livestock in this area [55], [56]. The specific method was to calculate the difference between the model-estimated number at the county scale and the actual statistical count. An adjustment coefficient was obtained, which was used to scale all grid values [43], [45]. The adjustment coefficient was calculated using the formula

$$A_i = P_i \times \frac{A_j}{P_j} \tag{7}$$

where i is the ith pixel, j is the administrative area of Menyuan County, A_i is the adjusted value of pixel i, P_i is the predicted value of pixel i, A_j is the statistical value of livestock in Menyuan County, and P_j is the total model-estimated livestock numbers in the county.

E. Analysis of TLCC and ALCC Spatiotemporal Dynamics

Using the optimal grassland AGB model constructed in previous period to simulate the annual maximum AGB in 2019–2023. The spatial distribution of TLCC from 2019 to 2023 was calculated using the spatial distribution of grassland AGB [(8)]. Regarding ALCC, only the spatial distribution from 2019 to 2022 was simulated due to a lack of data, such as land cover and population spatial distribution, in 2023

$$R_i = \frac{\mathrm{AGB}_i \times U}{I \times D} \tag{8}$$

where i is the ith pixel, R_i is the TLCC of pixel i, AGB_i is the annual maximum grassland AGB of pixel i estimated by the optimal model, I is the daily food intake of one sheep unit, D is the theoretical grazing days, and U is the reasonable utilization rate of grassland. Here, $I = 1.8$ kg and $D = 365$. It should be noted that the reasonable utilization rates vary among different grassland types. Due to alpine grassland accounts for over 95% of the total grassland area in Menyuan County, we applied a uniform utilization rate of 0.50 to simplify the TLCC calculation. All referenced from the Chinese Forestry and Grassland Industry Standard (NY/T 635–2015) [51].

The temporal dynamic changes of TLCC and ALCC were analyzed at the pixel level using the slope model [18]. For example, when slope > 0, TLCC tended to increase, and when slope < 0, TLCC tended to decrease. The F-test was further employed to determine the significance of TLCC change [significant if $F > F_{0.05}\ (1,\ n-2)$] [18]. According to the overlay analysis of slope and F-test, the temporal changes of TLCC could be divided into four levels: decrease, significant decrease, increase, and significant increase. The formula for the slope analysis is

$$\mathrm{slope} = \frac{n \times \sum_{i=1}^{n} i \times y_i - \sum_{i=1}^{n} i \sum_{i=1}^{n} y_i}{n \times \sum_{i=1}^{n} i^2 - \left(\sum_{i=1}^{n} i\right)^2} \tag{9}$$

where n is the time series length and y_i is the TLCC or ALCC of the ith year.

The formula for the F-test is

$$F = \frac{R^2(n-2)}{1-R^2} \tag{10}$$

where n is the time series length and R is the correlation coefficient between each pixel's TLCC or ALCC and the time series. Here, $F_{0.05}\ (1,\ 3) = 10.128$ for TLCC and $F_{0.05}\ (1,\ 2) = 18.513$ for ALCC.

F. Analysis of the Grassland–Livestock Balance

1) Calculation of Grassland–Livestock Balance Index: Combining TLCC and ALCC data from 2019 to 2022, the annual spatial distribution of the grassland–livestock balance index was analyzed using (11), and the grassland–livestock balance status of Menyuan County was quantitatively evaluated from 2019 to 2022. Notably, supplementary feeding was not considered in this study due to the complexity of feeding and the difficulty of obtaining relevant data

$$\mathrm{BGLI}_i = \frac{A_i - R_i}{R_i} \times 100\% \tag{11}$$

where i is the ith pixel, A_i is the ALCC of pixel i, R_i is the TLCC of pixel i, and BGLI_i is the grassland–livestock balance index of pixel i. The grassland–livestock status was considered nonoverloaded if $\mathrm{BGLI}_i \leq 0$, in grassland–livestock balance if $0 \leq \mathrm{BGLI}_i \leq 15\%$, overloaded if $15\% \leq \mathrm{BGLI}_i \leq 50\%$, and severely overloaded if $\mathrm{BGLI}_i \geq 50\%$. This standard was referenced from the Chinese Agriculture Industry Standard (LY/T 3320–2022) [4].

2) Analysis of Driving Factors on Grassland–Livestock Balance Dynamics: This study selected eight potential driving factors that may affect grassland–livestock balance dynamic from environmental and socioeconomic data, including elevation, slope, distance to river, precipitation, temperature, population, nighttime light, and livestock density. We further quantitatively explored the contribution of these factors on grassland–livestock balance change in Menyuan County using the LMG method [57], [58], [59]. The importance of driving

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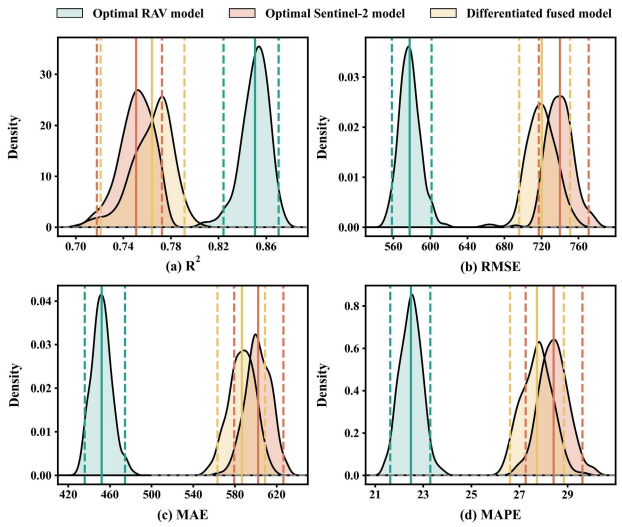


Fig. 2. Density distributions of (a) R^2, (b) RMSE, (c) MAE, and (d) MAPE for the grassland AGB models. The solid line indicates the average value, and the dashed line indicates the 95% confidence interval.

factors was normalized to achieve a total contribution rate of 100% [60]. A higher contribution rate indicates a more significant impact of the corresponding factor on the BGLI [61].

III. Results and Analysis

A. Variable Selection of the Grassland AGB Model

Both for the RAV data and Sentinel-2 data, all variable selection methods significantly reduced the number of variables. For RAV data, the number ranged from 9 to 13, while for Sentinel-2 data, it ranged from 9 to 11. Fig. 2 of the Supplementary Material presents the result of original variables selected by four methods and the R^2 between each selected variable and grassland AGB. For the RAV data, green, red edge, RGBVI, VDVI, and green (haralick correlation) were the variables selected most frequently by the various methods and were all selected by three methods simultaneously. For the Sentinel-2 data, the MCARI ($R^2 = 0.40$) with the highest determination coefficient of grassland AGB was selected simultaneously by four methods. The B2 (cluster progress), second in R^2 (0.30), was also retained by three methods. This indicated that variables with higher linear correlation with grassland AGB were more likely to be selected by different methods simultaneously.

B. Assessment of the Grassland AGB Model

1) RAV-Based Grassland AGB Model: RF, SVM, and ANN models were constructed using the RAV variables obtained from the four variable selection methods. Twelve grassland AGB estimation models were obtained, and the model accuracy evaluation result is shown in Table 7 of the Supplementary Material. The R^2 of the testing set indicated that the models' overall prediction accuracy was good, with six models reflecting more than 80% of the grassland AGB changes. The RF model constructed with LASSO-selected variables achieved the best prediction accuracy ($R^2 = 0.88$ and RMSE = 552.39 kg/ha).

2) Sentinel-2-Based Grassland AGB Model: RF, SVM, and ANN models were constructed using the Sentinel-2 variables obtained from the four selection methods, and 12 grassland AGB estimation models were obtained. The model accuracy evaluation result is shown in Table 8 of the Supplementary Material. The models' accuracy using Sentinel-2 variables was overall low (0.65–0.77) compared to those based on the RAV variables (0.70–0.88). The RF model constructed with SPA-selected variables was more robust than the others, with R^2 of 0.77 and RMSE of 692.41 kg/ha for the testing set.

3) Comparison Between the Optimal RAV Model, Optimal Sentinel-2 Model, and Differentiated Fused Model: Table I presents the various evaluation indicators of the optimal RAV model, optimal Sentinel-2 model, and differentiated fused model. The performance of the differentiated fused model was between the RAV model and the Sentinel-2 model. The accuracy of the differentiated fused model decreased by 9.09% compared to the optimal RAV model but was improved by 3.90% compared to the optimal Sentinel-2 model, with R^2 increasing from 0.77 to 0.80 and RMSE, MAE, and MAPE decreasing by 32.47 kg/ha, 9.94 kg/ha, and 0.57%, respectively. These results demonstrated that advantageous information in RAV spectra could be incorporated into Sentinel-2 spectral data through the differentiated fusion of spectral variables, effectively improving the potential of satellite-based grassland AGB estimation.

Based on the result of 200 repetitions of the optimal RAV model, optimal Sentinel-2 model, and differentiated fused model, the density distribution of the accuracy evaluation indicators of each model was obtained (Fig. 2). The result indicated that the optimal RAV model had the lowest difference and the highest accuracy in each iteration (mean values of R^2, RMSE, MAE, and MAPE were 0.85, 577.65 kg/ha, 452.10 kg/ha, and 22.48%, respectively). The stability of the optimal Sentinel-2 model and the differentiated fused model was not much different, but the latter was more robust (mean values of R^2, RMSE, MAE, and MAPE were 0.77, 720.24 kg/ha, 586.67 kg/ha, and 27.72%, respectively).

Fig. 3 shows the correlation and distribution between the model-estimated and the measured AGB. Although a few cases displayed overestimated low values and underestimated high values in prediction results of the differentiated fused model, most sample points clustered closely around the 1:1 line, indicating a generally good fit.

Considering spatial scale and prediction accuracy, the optimal RAV model was selected as the grassland AGB inversion model at the sample plot scale and the differentiated fused model was selected as the grassland AGB inversion model at the county scale.

C. Feasibility of Satellite-RAV Spectral Variable Fusion

The result of the consistency analysis of the two datasets (RAVSVs and MSISVs) (Table II) indicated that the correlation coefficients between the same variables ranged from 0.66 to 0.73, denoting a significant consistency between the two datasets. The correlation of RGBVI between RAV and Sentinel-2 was the highest, while the correlation of WDVI was the lowest. Fig. 4 presents the correlation of RAVSVs,

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TABLE I
ACCURACY EVALUATION OF THE OPTIMAL RAV MODEL, OPTIMAL SENTINEL-2 MODEL, AND DIFFERENTIATED FUSED MODEL

Model	Training dataset				Test dataset			
	R^2	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE
Optimal RAV model	0.98	221.96	167.87	8.37	0.88	552.39	437.86	21.85
Optimal Sentinel-2 model	0.97	259.14	194.56	9.86	0.77	692.41	563.73	26.88
Differentiated fused model	**0.98**	**228.19**	**188.31**	**9.56**	**0.80**	**659.94**	**553.79**	**26.31**

TABLE II
CORRELATION COEFFICIENTS OF SPECTRAL VARIABLES BETWEEN RAV AND SENTINEL-2

Spectral variable	Red edge	WDVI	EVI	RGBVI
correlation coefficient (R)	0.71*	0.66*	0.67*	0.73*

Note: * $p < 0.05$

TABLE III
ACCURACY EVALUATION OF THE LIVESTOCK DENSITY MODEL

Model	Training dataset		Test dataset	
	R^2	RMSE	R^2	RMSE
RF	**0.98**	**30.32**	**0.80**	**77.34**
SVM	0.55	135.28	0.53	128.34
ANN	0.85	75.58	0.67	101.19

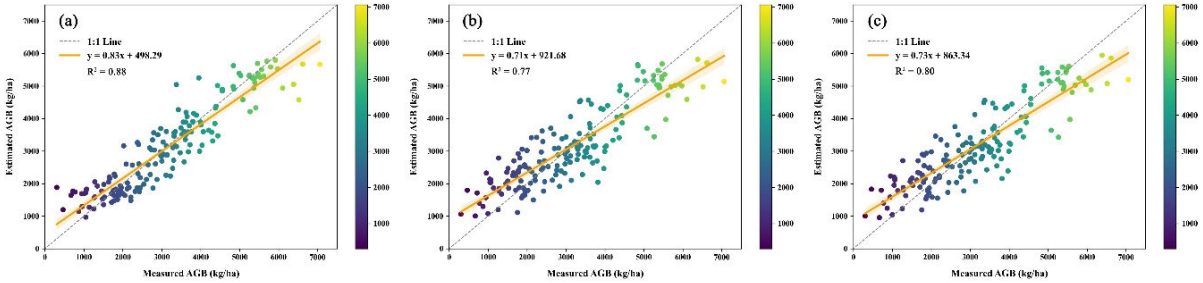


Fig. 3. Relationship between measured and estimated AGB. (a) Optimal RAV model, (b) optimal Sentinel-2 model, and (c) differentiated fused model.

MSISVs, and DFSVs with the grassland AGB. The correlation between DFSVs and AGB was greatly improved compared to that between MSISVs and grassland AGB, and it was close to that between RAVSVs and grassland AGB. The correlation coefficient between red edge and grassland AGB improved the most (+0.17) before and after fusion, while the correlation coefficient between RGBVI and grassland AGB improved the least (+0.06). The above results demonstrated that the NMF-based differentiated fusion method was suitable for spectral fusion between multispectral RAV and satellite data.

D. Assessment of the Livestock Density Model

The performances of livestock density models are presented in Table III. All three models could reflect over 50% of the changes in livestock density. The RF model exhibited the highest fitting performance and estimation accuracy, with a training set R^2 of up to 0.98 and a testing set R^2 and RMSE of 0.80 and 77.34 sheep/km^2, respectively. The ANN model had the second-best performance, with a testing set R^2 and RMSE of 0.67 and 101.19 sheep/km^2, respectively. The fitting degree and accuracy of the SVM model were the lowest.

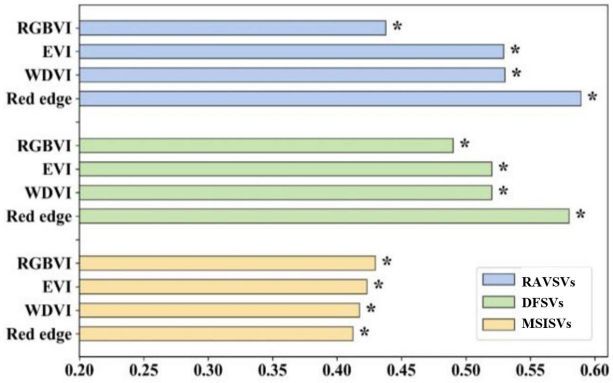


Fig. 4. Correlation between spectral variables and grassland AGB (* $p < 0.05$). RAVSVs: RAV spectral variables that satisfied the fusion condition; MSISVs: Sentinel-2 spectral variables that satisfied the fusion condition; DFSVs: fused spectral variables.

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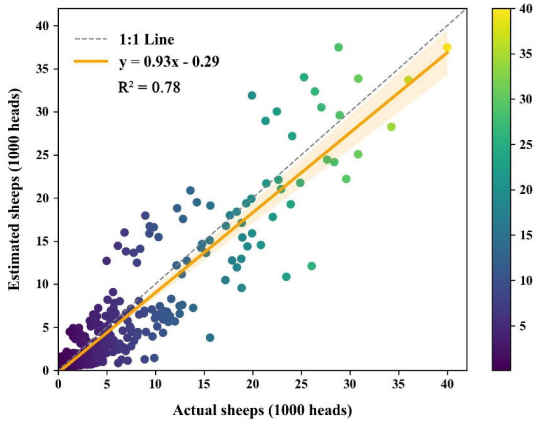


Fig. 5. Relationship between actual and estimated livestock numbers at the village scale.

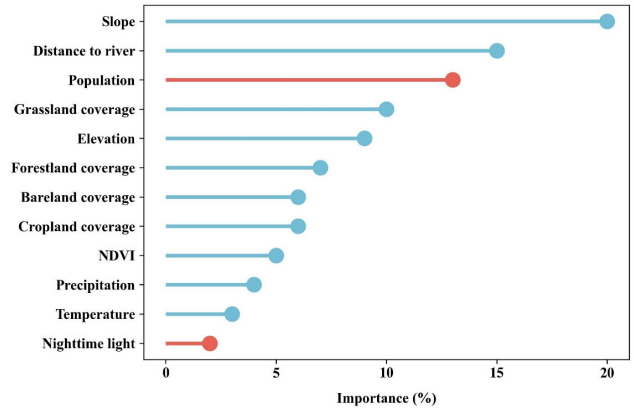


Fig. 6. Importance of factors in the optimal livestock density model. Blue: environmental factors; red: socioeconomic factors.

The grid-scale livestock density adjusted by the scaling coefficient was summarized zonally at the village scale and compared with the actual livestock statistical data within the village domain to evaluate the model's estimation performance at the grid scale (Fig. 5). The R^2 between model-estimated and actual livestock numbers reached 0.78, and the distribution characteristics of the two datasets were relatively consistent, indicating that the model was robust and reliably estimated livestock density at the grid scale.

In addition, the importance of each factor in the optimal livestock density model is shown in Fig. 6. The result indicated that slope, distance to river, population, and grassland coverage were four factors that best influenced the distribution of livestock, while NDVI, precipitation, temperature, and nighttime light had relatively small effects.

E. Spatiotemporal Dynamics of Grassland AGB, TLCC, and ALCC

1) Spatial Distribution of Grassland AGB: The spatial distribution of grassland AGB at the sample plot scale from 2021 to 2023 was obtained through inversion using the optimal RAV model and RAV images. Taking the inversion result of fixed sample area A on July 7, 2021, as an example (Fig. 7), grassland AGB values ranged from 864.68 to

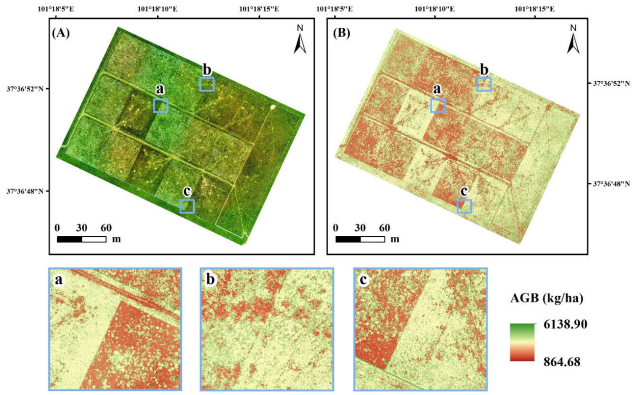


Fig. 7. Spatial distribution of grassland AGB at the fixed sample area A of the study area (July 7, 2021). (a) RGB image. (b) Inversion image.

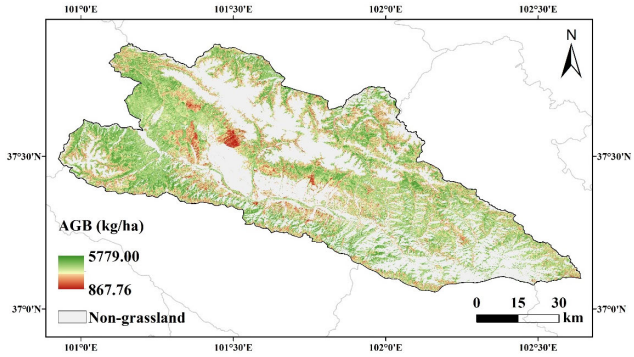


Fig. 8. Spatial distribution of the five-year average of annual maximum grassland AGB from 2019 to 2023 in the study area.

6138.90 kg/ha. The spatial pattern closely reflected the actual distribution of grassland AGB in the sample area. The above result indicated that, due to the detailed spatial resolution of RAV images, the sample-scale inversion yielded a high level of detail retention, and the multispectral RAV can realize excellent mapping accuracy.

The spatial distribution of the mean value of the annual maximum grassland AGB at the county scale from 2019 to 2023 was obtained through inversion based on the differentiated fused model and Sentinel-2 images (Fig. 8). The result indicated that the mean value of annual maximum grassland AGB in Menyuan County ranged from 867.76 to 5779.00 kg/ha, which was relatively consistent with the range of measured grassland AGB (304–7059.87 kg/ha). The overall distribution characteristics of grassland AGB were high in the periphery and low in the middle.

2) Spatiotemporal Dynamics of TLCC: The spatial distribution of the annual maximum TLCC for grassland from 2019 to 2023 in Menyuan County is presented in Fig. 9(a)–(e). The result indicated that the minimum TLCC value occurred in 2020 with 47 sheep/km^2 and the maximum in 2022 with 451 sheep/km^2. Combining the spatial distribution of TLCC over five years, areas with high TLCC were mainly distributed in the western and eastern parts. The western area had a large and continuous grassland area and a generally high TLCC, while grassland in the east was mostly separated by woodland and shrub, with a high degree of grassland fragmentation.

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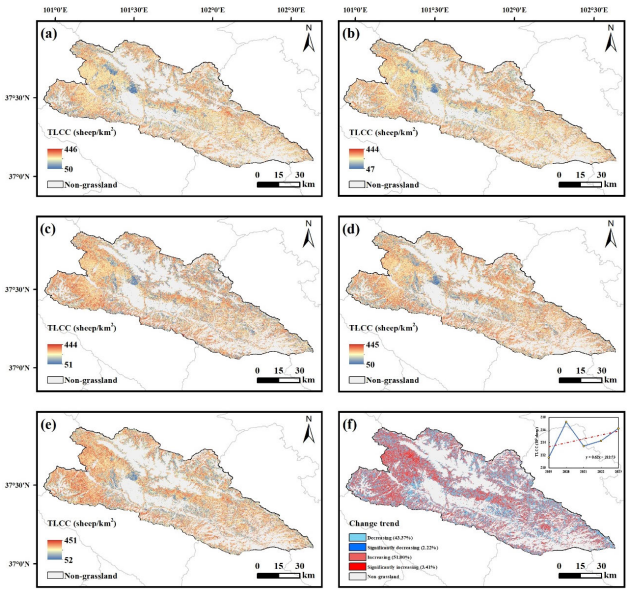


Fig. 9. Spatial distribution and change trend of annual maximum TLCC in the study area. (a) 2019, (b) 2020, (c) 2021, (d) 2022, (e) 2023, and (f) change trend from 2019 to 2023.

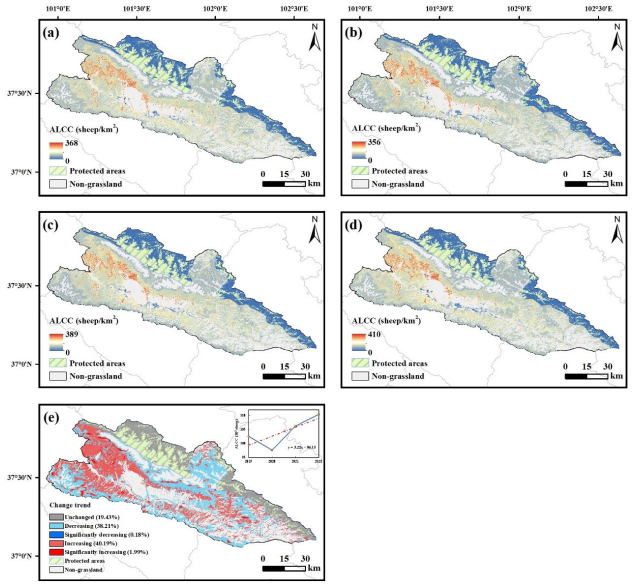


Fig. 10. Spatial distribution and change trend of ALCC in the study area. (a) 2019, (b) 2020, (c) 2021, (d) 2022, and (e) change trend from 2019 to 2022.

The high and low TLCC values exhibited a staggered distribution. The low TLCC in most central and southern regions is likely related to intensive human activities. The TLCC of the northern grassland was significantly affected by elevation, with higher elevation resulting in lower TLCC.

The change trend of grassland TLCC from 2019 to 2023 is presented in Fig. 9(f). Areas with a slightly increasing trend in TLCC covered 51.00% of the total grassland area, mainly concentrated in the southwestern part of Menyuan County, with sporadic distribution in eastern and central regions. Areas with a significant increase in TLCC represented 3.41% of the total grassland area, primarily located in the western part of Menyuan County. On the other hand, areas with a slight decreasing trend in TLCC occupied 43.37%, mainly in central and eastern Menyuan County. Areas with a significant decreasing trend in TLCC were 2.22%, concentrated in central Menyuan County. Notably, most grasslands with an increasing trend in TLCC had medium or high TLCC values, while most grasslands with a decreasing trend in TLCC had low TLCC values. Overall, the 2019–2023 grassland TLCC in Menyuan County was dominated by an increasing trend (54.41%) with an average annual growth rate of 0.62 sheep/km^2.

3) Spatiotemporal Dynamics of ALCC: The spatial distribution of the grassland ALCC from 2019 to 2022 in Menyuan County is presented in Fig. 10(a)–(d). Throughout the four years, the distribution pattern of ALCC remained consistent, with values ranging from 0 to 410 sheep/km^2. Overall, high ALCC values were predominantly observed in the western and central parts, with a limited presence observed in the eastern and southern regions. Areas with low ALCC were mainly found in northeastern parts of Menyuan County. This observation may be due to the fact that livestock prefer to congregate in areas with abundant grassland resources and good water and heat conditions. These resources provide good natural conditions for their survival, while ALCC is generally low in areas unsuitable for grazing, such as steep terrain, inaccessible transportation, and scarce water sources.

The change trend of grassland ALCC from 2019 to 2022 is presented in Fig. 10(e). The grassland area with a slight increase in ALCC accounted for 40.19% of the total grassland area and was mainly distributed in western Menyuan County, with small amounts in central and eastern parts. Areas with a significant increase in ALCC made up 1.99% and were mostly located in the northwest of Menyuan County. The percentage of grassland area with a slight decrease in ALCC was 38.21%, with a concentrated distribution in eastern and southwestern Menyuan County. Areas exhibited a significant decreasing trend in ALCC was very small, only 0.18%, and was scattered throughout the total area. Areas showed no change in ALCC (19.43%) were the core of the protected area that had been closed to grazing for many years, with a default ALCC of 0 at all times. Overall, the 2019–2022 grassland ALCC in Menyuan County was dominated by an increasing trend (42.18%), and the average annual growth rate was 3.22 sheep/km^2.

F. Grassland–Livestock Balance Status From 2019 to 2022

1) Spatiotemporal Dynamics of Grassland–Livestock Balance: The grassland–livestock balance status from 2019 to 2022 in Menyuan County is presented in Fig. 11(a)–(d). The grassland–livestock balance status in this region was good during the four years, with most areas consistently remaining at the nonoverloaded and balanced grades; however, the overloaded status was relatively serious in the core pasture area. The overall pattern of grassland–livestock balance was consistent over the four years. Nonoverloaded areas were mainly in the northern region of Menyuan County, with some distributed in high-altitude areas in the central and southern regions. Balanced areas were mainly concentrated in the west and east of Mengyuan County, while the areas of overloaded and severely overloaded status were mainly located in the

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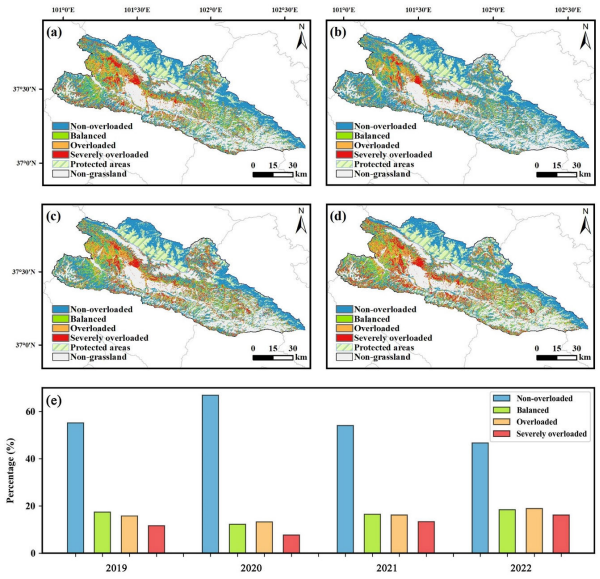


Fig. 11. Grassland–livestock balance status. (a) 2019, (b) 2020, (c) 2021, (d) 2022, and (e) change trend from 2019 to 2022.

pasture areas in the west and the periphery of high population density areas in the center. The distribution differences between different years were mainly reflected in the extent of overloaded grassland, which was the smallest in 2020 and expanded significantly by 2022. Notably, despite a high TLCC, some regions were still overloaded because of high ALCC.

Fig. 11(e) illustrates the interannual variation of grassland–livestock balance grades. Over the four years, the area of nonoverloaded grassland was consistently the largest, while the area of severely overloaded grassland was consistently the smallest. The percentage of grassland area that was nonoverloaded and balanced was 72.54% in 2019, increased to 79.10% in 2020, and began to decrease to 70.48% and 65.00% in 2021 and 2022, respectively. The interannual change trends of the four grades indicated that the nonoverloaded grassland area increased first and then decreased, while the grassland area of the other three grades decreased first and then increased. Additionally, the area of nonoverloaded and severely overloaded grassland fluctuated greatly between years, while the area of balanced and overloaded grassland fluctuated little.

2) Accuracy Validation of Grassland–Livestock Balance: To evaluate the mapping accuracy of the grassland–livestock balance, we selected five areas with different grassland–livestock balance grades for further analysis. The number of livestock, vector boundary, and grazing area of each fenced plot were recorded accurately in field investigation. The spatial distributions of the grassland–livestock balance grades in these plots are shown in Fig. 12(a)–(e). By comparing the RGB images of Sentinel-2, it can be seen that the inversion result of each plot was basically consistent with the true grassland–livestock balance grade, and the differences between different grassland–livestock balance grades were obvious. The accuracy of each area ranged from 62.47% to 85.40% [Fig. 12(f)]. Despite some salt-and-pepper noise in the classification result, the average accuracy of the five areas

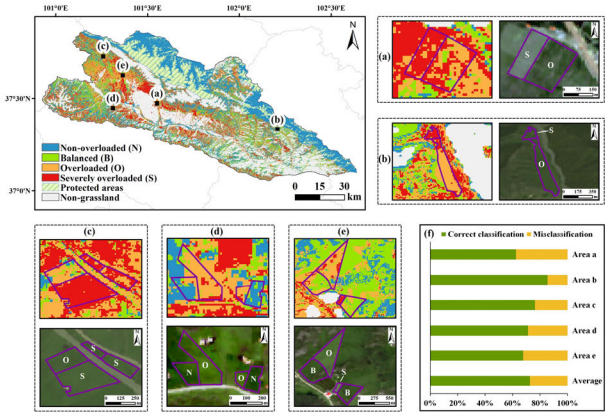


Fig. 12. Accuracy evaluation of grassland–livestock balance in five areas. (a)–(e) Represent different evaluation areas respectively. (f) Represents the classification accuracy of each area.

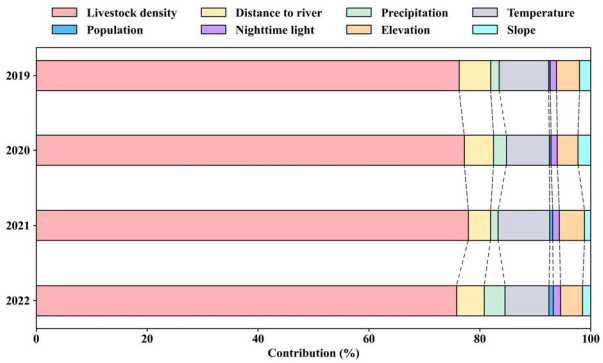


Fig. 13. Contribution rate of each driving factor on the dynamics of grassland–livestock balance.

reached 72.63%, indicating that this method can effectively identify the grassland–livestock balance status.

3) Contribution of Driving Factors on Grassland–Livestock Dynamics: Fig. 13 shows the contribution rate of each driving factor to the change of grassland–livestock balance based on the LMG method. The contribution rate of each factor remained relatively stable across different years. Notably, livestock density contributed over 75% to the spatiotemporal change of grassland–livestock balance, playing an absolutely dominant role. This contribution showed a trend of increasing and then decreasing from 2019 to 2022. The contributions of temperature, distance to river, elevation, precipitation, and slope ranked from second to sixth, with average values of 8.47%, 4.98%, 4.09%, 2.24%, and 1.74%, respectively. The contribution rates of these factors all fluctuated lightly throughout 2019–2022. The effects of nighttime light and population on the grassland–livestock balance were weak but showed a gradually increasing trend.

IV. Discussion

A. Spectral Variables for Grassland AGB Estimation

In this study, the spectral variables selected for the optimal RAV model were green, red edge, WDVI, EVI, and RGBVI, and the spectral variables selected for the optimal Sentinel-2 model were B6 (red edge2), MCARI, SATVI, WDVI, EVI,

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RGBVI, and S2REP (Fig. 2 of the Supplementary Material). By analyzing the R^2 between each spectral variable and grassland AGB, the red edge band displayed excellent performance in quantifying grassland AGB. Such good performance is often attributed to the red edge band's sensitivity to plant chlorophyll content and changes; additionally, it effectively reduces the spectral saturation effect in high biomass regions [13], [14], [15]. The visible light and NIR band also made important contributions to estimating grassland. Kong et al. [62] monitored the alpine grassland AGB in the northern Tibetan Plateau based on hyperspectral ground spectrometer and HJ-1A data and observed that the vegetation indices constructed using green, red, and NIR bands had a good correlation with grassland AGB. In this study, only one vegetation index (SATVI) was calculated using SWIR band, was selected by the optimal model, and contributed relatively little to grassland AGB estimation. Ji et al. [48] estimated the grassland AGB in east-central Mongolia based on various optical remote sensing data and observed that the SWIR band was only moderately important, which was consistent with the conclusion of this study. This result may be due to the field observations in this study being concentrated in grassland growing season, during which the majority of alpine grassland was photosynthetic vegetation. However, some studies have found that vegetation indices constructed based on the SWIR band have a good estimation performance for nonphotosynthetic vegetation in grassland [35], [63]. Therefore, in the future estimation of grassland AGB, selecting more suitable vegetation indices according to different grassland growth periods could be considered.

B. Texture Variables for Grassland AGB Estimation

In addition to spectral variables, texture variables were essential in estimating grassland AGB. Unlike vegetation indices, which are computed pixel by pixel, texture variables are computed based on the local neighborhood of pixels, maximizing their ability to estimate grassland AGB by adjusting the neighborhood's size [9]. Many studies have demonstrated that spectral variables were prone to saturation in high biomass areas, limiting their effectiveness in estimating high biomass [64]. Texture can effectively depict the structural information of vegetation, and including texture variables can effectively improve the estimation of high biomass areas [65]. Kelsey and Neff [9] believed that the excellent performance of texture variables in predicting vegetation biomass could also be attributed to window aggregation in texture analysis, as this process effectively reduced the errors generated by field sampling and image geographic calibration. In this study, half of the variables used in the optimal RAV model were texture variables, whereas texture variables accounted for only 4/11 of the optimal Sentinel-2 model variables. The coefficients of determination of the RAV texture variables with the grassland AGB were also significantly higher than those of the Sentinel-2 texture variables. The above results indicate that the effectiveness of texture analysis depends, to some extent, on the spatial resolution of remote sensing images [50]. Additionally, texture features in green, red, and NIR bands were more important in grassland AGB estimation, as highlighted by Gallardo-Cruz et al. [66].

C. Differentiated Fusion Method

In this study, the RAV-based models act as the benchmark, which provide detailed AGB maps for specific locations; the Sentinel-2-based models provide regional coverage with a good balance between spatial and temporal resolution; and the fused model aims to reduce the accuracy gap between the two by transferring advantageous information from the RAV data to the Sentinel-2 data.

Differentiated fusion of RAV and Sentinel-2 spectral variables was realized using the NMF algorithm. The effect of differentiated fusion depends heavily on the original matrices and is additionally sensitive to some parameters (e.g., factorization factor and loss function [67]). Original matrices are original remote sensing data from different platforms, and their accuracy can be improved by preprocessing steps, such as removing outliers [68]. The factorization factor number is the most important parameter; it directly determines the result of the matrix decomposition. Generally, the larger the number of factorization factors, the closer the decomposition result is to the original matrix. However, the model's complexity increases with the number of factorization factors, and excessive complexity can lead to the model's overfitting [69]. Additionally, NMF cannot handle data's noise during the decomposition process, so if the number of factorization factors is too big, the noise will also be decomposed into the matrix [27], [35]. We determined the optimal factorization factor number using PSNR, which is more sensitive to noise than other evaluation indicators [70]. The loss function is another important parameter in NMF, and we measured the error of the matrix decomposition using the Frobenius norm. Compared to other loss functions, the Frobenius norm is more robust to noise and outliers and can improve the accuracy of the reconstruction result [71].

Compared to traditional linear spectrum unmixing, NMF can obtain higher spectral retention and spatial fidelity [72]. However, so far, almost all NMF-based differentiated fusion studies have been realized using ground-based hyperspectral and satellite data, and the high cost and cumbersome processing of hyperspectral data pose challenges for large-scale and long-term spectral data fusion. Additionally, the NMF-based differentiated fusion method is still in the preliminary exploration stage of grassland AGB estimation. In this study, the NMF-based differentiated fusion was innovatively adopted to realize the fusion between multispectral RAV and satellite data, and the high-precision inversion of grassland AGB was well realized. The result indicated that the accuracy of the fused Sentinel-2 variables in estimating grassland AGB improved by 3.9%.

However, only conventional methods were used for preprocessing the original data in this study, which may not fully address noise, especially in the field-collected RAV data. Lv et al. [73] proposed an iterative adaptive spatial filter that performed well in smoothing noise of hyperspectral remote sensing data. The advanced denoising algorithms can

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be considered in our future research to further reduce the influence of noise in the process of differentiated fusion.

D. Importance of Factors for Livestock Density Spatialization

In this study, slope, distance to river, population, and grassland coverage strongly influenced livestock spatial distribution, consistent with the findings of Li et al. [44]. Generally, the slope is an important factor affecting livestock accessibility, and areas with high slopes tend to be steep and inaccessible, which is not conducive to livestock movement and feeding. Distance to the river is highly correlated with the natural resources required for livestock to survive; areas near the river often have sufficient water sources and lush vegetation, which is more conducive to livestock drinking and feeding. The number and distribution of livestock are also influenced by human grazing behavior, with grazing areas usually expanding outward from human activity areas, resulting in relatively high livestock density in grassland near densely populated areas. Grassland coverage reflects the amount of forage available for livestock feeding to some extent; it is a key factor influencing livestock movement and distribution, and livestock density is generally high in areas with high grassland coverage. The result of Wu's [74] spatialization of ALCC in Qinghai Province, China, indicated that precipitation and temperature largely impacted the spatial distribution of livestock; however, this was not reflected in our study, probably because Menyuan County is not large and has small latitudinal and longitudinal spans. Therefore, the spatial differences in precipitation and temperature within the county were not significant, which led to the relatively low importance of the two parameters in this study's model.

E. Strengths and Limitations of Grassland–Livestock Balance Analysis

In this study, we created high-precision grassland TLCC and ALCC raster datasets for Menyuan County from 2019 to 2022, realizing a fine and continuous assessment of the annual distribution dynamics of grassland–livestock balance. Since the estimations of TLCC and ALCC were achieved through robust machine learning algorithms, the practice of extending the models constructed using multiyear samples to adjacent years is widely recognized. Besides, the analysis of potential driving factors on grassland–livestock balance dynamics also indicated that the contribution rate of each factor remained relatively stable across different years. This suggested that the main driving factors had a consistent effect on the grassland–livestock balance. Therefore, this assessment framework could be extrapolated to a longer period. However, the construction of the TLCC and ALCC datasets still can be improved. In terms of TLCC: 1) although our sample plots were distributed as uniformly as possible to capture diverse grassland conditions in the study area, extrapolating the model developed using plot-scale data to the regional scale is a well-established approach [75], [76], and differences between sample and nonsample areas cannot be completely avoided. Therefore, expanding the number and spatial distribution of sample plots to construct a more robust model is advisable. Besides, verifying the model performance using independent samples from nonsample areas is also an effective way to reduce differences; 2) the optimal AGB model slightly overestimated the low values and underestimated the high values, indicating that they had more potential uncertainties in areas with particularly low or high AGB. Thus, it is necessary to supplement more extreme value samples that tend to be misestimated in future survey; and 3) the AGB estimation models were constructed using conventional machine learning algorithms due to the limited samples. In subsequent studies, the utilization of deep learning or more advanced algorithms can be attempted to further improve the model accuracy. In terms of ALCC: 1) we considered various types of environmental and socioeconomic factors that impacted the livestock density distribution; they can reflect the spatial distribution of livestock well, but some factors may have been overlooked or data-constrained and not included; 2) livestock statistical data were collected and organized by the county government, and the errors would inevitably arise during the statistical process of reporting from herders to each level of government; additionally, due to the extreme difficulty in obtaining accurate statistical data at the herder level, only village-scale livestock statistical data were modeled, which affected the accuracy of ALCC spatialization to some extent; and 3) limited by data availability, the coarse spatial resolution of the individual environmental and socioeconomic data reduced the fineness of ALCC spatialization, affecting the spatial match with the TLCC dataset at the 10-m resolution. However, with continuous updating and optimization of the datasets, the above limitations can be subsequently reduced by replacing and using data with higher spatiotemporal resolution. In addition, the follow-up plan may consider using physical guided methods to better develop the research on the spatialization of livestock statistical data.

Moreover, this study has some limitations in assessing grassland–livestock balance. On the one hand, the effect of supplementary feeding on the analysis of grassland–livestock balance has not been considered. Supplementary feeding usually occurs in winter and spring when grassland AGB is scarce. Under supplementary feeding, the livestock will naturally reduce the foraging requirement for natural grassland, which will affect the equilibrium relationship between TLCC and ALCC. However, the field survey period of this study was from July to September, which is the peak growing season of alpine grassland. The grassland AGB usually peaked during this period, and the TLCC calculated using the annual maximum grassland AGB could be approximated as the maximum TLCC in a year. Although supplementary feeding behavior rarely occurs during the peak growing season of grassland, it may still occur in small amounts in areas with high utilization, severe degradation, and low forage production. Therefore, adding supplementary feeding data to the existing system can further improve the accuracy and practical value of grassland–livestock balance assessment results; at the same time, attempts can be made to expand the study of grassland–livestock balance outside the grassland growing season. However, accurate supplementary feeding data are extremely difficult to obtain because it requires household

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surveys, which are difficult, costly, and laborious. According to previous surveys, we found that some herders did not cooperate well with the interviewers, probably due to the influence of the grassland ecology's compensatory policy and other factors. Additionally, regarding the close structure and number of livestock, the supplementary feeding data provided by different herders varied greatly, the authenticity and accuracy of the data could not be considered. In the future, we could cooperate with the local grassland station through data sharing to formulate a detailed and feasible survey plan for supplementary feeding. With the coordination and assistance of government staff, a combination of household and questionnaire surveys could be used to systematically record the methods and amounts of supplementary feeding for livestock.

On the other hand, the applicability of the models constructed at the county scale to estimate the TLCC and ALCC of grassland in larger regions needs to be further verified. The specific verification process is to directly use the model frameworks constructed in this study to obtain the TLCC and ALCC spatial distributions of the target area by inversion and verify the accuracy of the inversion results using local grassland field data and village-level livestock statistical data. Generally speaking, the more similar the climate condition, grassland type and grazing pattern of the target area are to Menyuan County, the higher the transferability of TLCC and ALCC assessments. Besides, combining local samples from the target area to reconstruct and optimize existing models is an effective way to improve models' transferability. In the future, we will conduct additional grassland–livestock balance surveys at a larger regional scale to verify the generalizability of the models established in this study, and we plan to combine more measured sample data and strive to extrapolate the research scale to a larger region.

F. Ecological Effects and Practical Applications of Grassland–Livestock Balance

It is crucial to consider the long-term effects on the environment, ecosystem, and climate based on the result of grassland–livestock balance. The balance state between grassland and livestock is the basis for maintaining the health and sustainable development of grassland ecosystems. Regardless of which direction this balance tilts toward, it is detrimental. When TLCC $>$ ALCC, it indicates underutilization of grassland resources, which can hinder the grassland's natural renewal and regeneration processes. For example, long-term grazing exclusion can lead to excessive litter accumulation on the grassland surface. This can hinder vegetation photosynthesis by reducing light penetration and also impact microbial respiration by altering soil conditions [77], [78]. Finally, the grassland ecosystem may exhibit characteristics of an "aged" state and lead to a decline in overall productivity and resilience. This state can even influence local microclimates, affecting temperature regulation and water circulation [79], [80]. Furthermore, the failure to maximize the grassland productivity can also constrain the livestock production and economic development. Conversely, when TLCC $<$ ALCC, the adverse effects are obvious. Overgrazing is widely recognized as a primary factor of grassland degradation [81]. Excessive grazing pressure disrupts the delicate balance between plant growth and consumption, hindering the accumulation of organic matter and essential nutrients in the soil [82]. Additionally, over utilization reduces plant diversity and facilitates the invasion of toxic weeds [83], [84]. Soil erosion becomes a significant concern with the reduction of protective vegetation, leaving the soil vulnerable to wind and water. The reduced vegetation and soil erosion also weaken the carbon sequestration capacity of grassland, potentially further exacerbating climate change [85]. Meanwhile, the reduction of grassland productivity makes it challenging for the ecosystem to support the existing livestock population and leading to economic losses for pastoralists. The healthy grassland–livestock balance status contributes to grassland recovery and further improves the grassland ecological environment; and the improved ecological environment means better climate conditions for grasslands, which in turn further promotes grassland restoration [86].

The assessment results of the grassland–livestock balance in this study can provide theoretical guidance to adjust or optimize grazing decision-making in time and assist governments in mitigating risks that grassland ecosystems may face. This includes: 1) grazing strategies: implementing seasonal grazing exclusion and rotational grazing in overloaded areas (e.g., northwestern Menyuan County), or transferring some livestock to nonoverloaded areas (e.g., southeastern Menyuan County) to fully utilize grassland resources; 2) ecological migration policies: encouraging migration from overloaded areas (e.g., core pasture area) to reduce human impact and facilitate ecological restoration; and 3) supplementary feeding strategies: implementing targeted feeding strategies, particularly in areas with limited forage production, to reduce the pressure on natural grasslands.

V. Conclusion

In this study, we proposed a differentiated fusion method of satellite-RAV data, combined with various variable selection methods and machine learning algorithms to create a 10-m spatial resolution TLCC dataset for Menyuan County, Qinghai Province, from 2019 to 2023. Additionally, a scheme for spatial decomposition of livestock statistical data of the smallest administrative unit using gridded environmental and socioeconomic data was proposed to create a high-precision ALCC dataset for this region from 2019 to 2022. On this basis, the annual grassland–livestock balance status of the study area from 2019 to 2022 was quantitatively explored through the grassland–livestock balance index. The main conclusions are as follows: 1) both spectral and texture features played important roles in estimating grassland AGB, with red edge and visible light being the key bands in spectral features and NIR and visible light in texture features; 2) the optimal models constructed by RF algorithm at the sample plot and county scales reflected 88% and 77% of the change in grassland AGB, respectively; 3) the NMF-based differentiated fusion method improved the correlation coefficients of Sentinel-2 spectral variables with grassland AGB by more than 14%, effectively enhancing the ability of estimating grassland AGB

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at the regional scale by Sentinel-2 (R^2 from 0.77 to 0.80); 4) the livestock density model constructed by the RF algorithm had the best spatialization effect of ALCC ($R^2 = 0.80$ and RMSE = 77.34 sheep/km^2); 5) the classification accuracy of grassland–livestock balance reached 72.63%, which can effectively distinguish the grassland–livestock balance grades. The overall grassland–livestock balance status in Menyuan County was good from 2019 to 2022, with more than 65% of grassland consistently at nonoverloaded and balanced grades; overloaded and severely overloaded grassland was predominantly found in the west and center; the area of nonoverloaded and severely overloaded grassland had large interannual fluctuations; and 6) the livestock density played an absolute dominant role in grassland–livestock balance dynamics.

However, this study has not considered the effect of supplementary feeding on the analysis of grassland–livestock balance. In the future, we will systematically collect supplementary feeding data to further improve the accuracy and practical value of grassland–livestock balance assessment.

REFERENCES

[1] Y. Wang et al., "Grassland changes and adaptive management on the Qinghai–Tibetan Plateau," *Nature Rev. Earth Environ.*, vol. 3, no. 10, pp. 668–683, Sep. 2022, doi: 10.1038/s43017-022-00330-8.

[2] Q. Qian et al., "Improving herders' income through Alpine grassland husbandry on Qinghai-Tibetan Plateau," *Land Use Policy*, vol. 113, Feb. 2022, Art. no. 105896, doi: 10.1016/j.landusepol.2021.105896.

[3] M. Yang et al., "Trade-offs in ecological, productivity and livelihood dimensions inform sustainable grassland management: Case study from the Qinghai-Tibetan Plateau," *Agricult., Ecosyst. Environ.*, vol. 313, Jun. 2021, Art. no. 107377, doi: 10.1016/j.agee.2021.107377.

[4] *Evaluating Criterion for Balance of Forage Supply and Livestock Requirement*, document LY/T 3320-2022, National Forestry and Grassland Administration and Ministry of Natural Resources of the People's Republic of China, Beijing, 2022.

[5] J. Zhang, L. Zhang, W. Liu, Y. Qi, and X. Wo, "Livestock-carrying capacity and overgrazing status of Alpine grassland in the Three-River Headwaters region, China," *J. Geographical Sci.*, vol. 24, no. 2, pp. 303–312, Apr. 2014, doi: 10.1007/s11442-014-1089-z.

[6] E. Grüner, M. Wachendorf, and T. Astor, "The potential of UAV-borne spectral and textural information for predicting aboveground biomass and n fixation in legume-grass mixtures," *PLoS ONE*, vol. 15, no. 6, Jun. 2020, Art. no. e0234703, doi: 10.1371/journal.pone.0234703.

[7] C. A. Coburn and A. C. B. Roberts, "A multiscale texture analysis procedure for improved forest stand classification," *Int. J. Remote Sens.*, vol. 25, no. 20, pp. 4287–4308, Oct. 2004, doi: 10.1080/0143116042000192367.

[8] D. Lu, "Aboveground biomass estimation using Landsat TM data in the Brazilian Amazon," *Int. J. Remote Sens.*, vol. 26, no. 12, pp. 2509–2525, Jun. 2005, doi: 10.1080/01431160500142145.

[9] K. Kelsey and J. Neff, "Estimates of aboveground biomass from texture analysis of Landsat imagery," *Remote Sens.*, vol. 6, no. 7, pp. 6407–6422, Jul. 2014, doi: 10.3390/rs6076407.

[10] J. Franke, V. Keuck, and F. Siegert, "Assessment of grassland use intensity by remote sensing to support conservation schemes," *J. Nature Conservation*, vol. 20, no. 3, pp. 125–134, Jun. 2012, doi: 10.1016/j.jnc.2012.02.001.

[11] G. Lei et al., "The quantitative estimation of grazing intensity on the Zoige Plateau based on the space-air-ground integrated monitoring technology," *Remote Sens.*, vol. 12, no. 9, p. 1399, Apr. 2020, doi: 10.3390/rs12091399.

[12] M. Lange, H. Feilhauer, I. Kühn, and D. Doktor, "Mapping land-use intensity of grasslands in Germany with machine learning and Sentinel-2 time series," *Remote Sens. Environ.*, vol. 277, Aug. 2022, Art. no. 112888, doi: 10.1016/j.rse.2022.112888.

[13] M. Guerini Filho, T. M. Kuplich, and F. L. F. D. Quadros, "Estimating natural grassland biomass by vegetation indices using Sentinel 2 remote sensing data," *Int. J. Remote Sens.*, vol. 41, no. 8, pp. 2861–2876, Apr. 2020, doi: 10.1080/01431161.2019.1697004.

[14] X. Jiang, S. Fang, X. Huang, Y. Liu, and L. Guo, "Rice mapping and growth monitoring based on time series GF-6 images and red-edge bands," *Remote Sens.*, vol. 13, no. 4, p. 579, Feb. 2021, doi: 10.3390/rs13040579.

[15] Y. Kang, Q. Meng, M. Liu, Y. Zou, and X. Wang, "Crop classification based on red edge features analysis of GF-6 WFV data," *Sensors*, vol. 21, no. 13, p. 4328, Jun. 2021, doi: 10.3390/s21134328.

[16] Y. Qin et al., "Effects of Plateau Pikas' foraging and burrowing activities on vegetation biomass and soil organic carbon of Alpine grasslands," *Plant Soil*, vol. 458, nos. 1–2, pp. 201–216, Jan. 2021, doi: 10.1007/s11104-020-04489-1.

[17] X. Lyu, X. Li, D. Dang, H. Dou, K. Wang, and A. Lou, "Unmanned aerial vehicle (UAV) remote sensing in grassland ecosystem monitoring: A systematic review," *Remote Sens.*, vol. 14, no. 5, p. 1096, Feb. 2022, doi: 10.3390/rs14051096.

[18] J. Ge et al., "Modeling Alpine grassland cover based on MODIS data and support vector machine regression in the headwater region of the Huanghe river, China," *Remote Sens. Environ.*, vol. 218, pp. 162–173, Dec. 2018, doi: 10.1016/j.rse.2018.09.019.

[19] C. I. Alvarez-Mendoza et al., "Predictive modeling of above-ground biomass in Brachiaria pastures from satellite and UAV imagery using machine learning approaches," *Remote Sens.*, vol. 14, no. 22, p. 5870, Nov. 2022, doi: 10.3390/rs14225870.

[20] T. G. Morais, R. F. M. Teixeira, M. Figueiredo, and T. Domingos, "The use of machine learning methods to estimate aboveground biomass of grasslands: A review," *Ecolog. Indicators*, vol. 130, Nov. 2021, Art. no. 108081, doi: 10.1016/j.ecolind.2021.108081.

[21] C. Pohl and J. L. Van Genderen, "Review article multisensor image fusion in remote sensing: Concepts, methods and applications," *Int. J. Remote Sens.*, vol. 19, no. 5, pp. 823–854, Jan. 1998, doi: 10.1080/014311698215748.

[22] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017, doi: 10.1109/LGRS.2017.2704625.

[23] Y. Yang, W. Wan, S. Huang, F. Yuan, S. Yang, and Y. Que, "Remote sensing image fusion based on adaptive IHS and multi-scale guided filter," *IEEE Access*, vol. 4, pp. 4573–4582, 2016, doi: 10.1109/ACCESS.2016.2599403.v.

[24] A. Azarang, H. E. Manoochehri, and N. Kehtarnavaz, "Convolutional autoencoder-based multispectral image fusion," *IEEE Access*, vol. 7, pp. 35673–35683, 2019, doi: 10.1109/ACCESS.2019.2905511.

[25] Z. Lv, P. Zhang, W. Sun, J. A. Benediktsson, and T. Lei, "Novel land-cover classification approach with nonparametric sample augmentation for hyperspectral remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407613, doi: 10.1109/TGRS.2023.3309949.

[26] Z. Lv, J. Liu, W. Sun, T. Lei, J. A. Benediktsson, and X. Jia, "Hierarchical attention feature fusion-based network for land cover change detection with homogeneous and heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411115, doi: 10.1109/TGRS.2023.3334521.

[27] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022, doi: 10.1109/JPROC.2022.3219376.

[28] C. He, B. Gao, Q. Huang, Q. Ma, and Y. Dou, "Environmental degradation in the urban areas of China: Evidence from multi-source remote sensing data," *Remote Sens. Environ.*, vol. 193, pp. 65–75, May 2017, doi: 10.1016/j.rse.2017.02.027.

[29] Y. Ma, H. Chen, G. Zhao, Z. Wang, and D. Wang, "Spectral index fusion for salinized soil salinity inversion using Sentinel-2A and UAV images in a coastal area," *IEEE Access*, vol. 8, pp. 159595–159608, 2020, doi: 10.1109/ACCESS.2020.3020325.

[30] B. Ren, L. Pueyo, G. B. Zhu, J. Debes, and G. Duchêne, "Non-negative matrix factorization: Robust extraction of extended structures," *Astrophysical J.*, vol. 852, no. 2, p. 104, Jan. 2018, doi: 10.3847/1538-4357/aaa1f2.

[31] J. Gan, T. Liu, L. Li, and J. Zhang, "Non-negative matrix factorization: A survey," *Comput. J.*, vol. 64, no. 7, pp. 1080–1092, Aug. 2021, doi: 10.1093/comjnl/bxab103.

[32] F. Z. Benhalouche, M. S. Karoui, Y. Deville, and A. Ouamri, "Hyperspectral and multispectral data fusion based on linear-quadratic nonnegative matrix factorization," *J. Appl. Remote Sens.*, vol. 11, no. 2, May 2017, Art. no. 025008, doi: 10.1117/1.jrs.11.025008.

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[33] F. Yang, F. Ma, S. Huo, and Y. Wang, "Multi-sensor data fusion of remotely-sensed images with sparse and logarithmic low-rank regularization for shadow removal and denoising," *Int. J. Remote Sens.*, vol. 42, no. 18, pp. 6961–6983, Sep. 2021, doi: 10.1080/01431161.2021.1941388.

[34] H. Chen, Y. Ma, A. Zhu, Z. Wang, G. Zhao, and Y. Wei, "Soil salinity inversion based on differentiated fusion of satellite image and ground spectra," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102360, doi: 10.1016/j.jag.2021.102360.

[35] R. Guo et al., "Estimating aboveground biomass of Alpine grassland during the wilting period using in situ hyperspectral, Sentinel-2, and Sentinel-1 data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4400516, doi: 10.1109/TGRS.2023.3341956.

[36] Y. Qu, Y. Zhao, G. Ding, W. Chi, and G. Gao, "Spatiotemporal patterns of the forage-livestock balance in the Xilin Gol steppe, China: Implications for sustainably utilizing grassland-ecosystem services," *J. Arid Land*, vol. 13, no. 2, pp. 135–151, Feb. 2021, doi: 10.1007/s40333-021-0053-x.

[37] L. Ma et al., "Monitoring and influencing factors of grassland livestock overload in Xinjiang from 1982 to 2020," *Frontiers Plant Sci.*, vol. 15, Mar. 2024, Art. no. 1340566, doi: 10.3389/fpls.2024.1340566.

[38] X. Guo, W. Liu, C, Meng, S. Hu, S. Liu, and Z. Lin, "Variations of forage yield and forage-livestock balance in grasslands over the Tibetan-Plateau, China," *Chin. J. Appl. Ecology*, vol. 32, no. 7, pp. 2415–2425, Jul. 2021, doi: 10.13287/j.1001-9332.202107.002.

[39] N. Dong, X. Yang, and H. Cai, "Research progress and perspective on the spatialization of population data," *J. Geo-Inf. Sci.*, vol. 18, no. 10, pp. 1295–1304, Oct. 2016.

[40] M. F. Goodchild, L. Anselin, and U. Deichmann, "A framework for the areal interpolation of socioeconomic data," *Environ. Planning A, Economy Space*, vol. 25, no. 3, pp. 383–397, Mar. 1993, doi: 10.1068/a250383.

[41] K. Neumann et al., "Modelling the spatial distribution of livestock in Europe," *Landscape Ecol.*, vol. 24, no. 9, pp. 1207–1222, Nov. 2009, doi: 10.1007/s10980-009-9357-5.

[42] D. J. Prosser et al., "Modelling the distribution of chickens, ducks, and geese in China," *Agricult., Ecosystems Environ.*, vol. 141, nos. 3–4, pp. 381–389, May 2011, doi: 10.1016/j.agee.2011.04.002.

[43] T. P. Robinson et al., "Mapping the global distribution of livestock," *PLoS ONE*, vol. 9, no. 5, May 2014, Art. no. e96084, doi: 10.1371/journal.pone.0096084.

[44] X. Li, J. Hou, and C. Huang, "High-resolution gridded livestock projection for western China based on machine learning," *Remote Sens.*, vol. 13, no. 24, p. 5038, Dec. 2021, doi: 10.3390/rs13245038.

[45] M. Gilbert et al., "Global distribution data for cattle, buffaloes, horses, sheep, goats, pigs, chickens and ducks in 2010," *Sci. Data*, vol. 5, no. 1, Oct. 2018, Art. no. 180227, doi: 10.1038/sdata.2018.227.

[46] Y. Yan, Q. Ning, Z. Wang, and Q. Dou, "Analysis of highland barley phenophase change trend and the driving factors in Menyuan County, Qinghai province during 1980–2015," *Acta Ecologica Sinica*, vol. 38, no. 4, pp. 1–8, Feb. 2018, doi: 10.5846/stxb201701090067.

[47] *Technical Specification for Investigation and Assessment of National Ecological Status*, document HJ 1168-2021, Ministry of Ecology and Environment of the People's Republic of China, Beijing, 2021.

[48] S. Ji, B. Dashpurev, T. N. Phan, M. Dorj, Y. Jäschke, and L. Lehnert, "Above-ground biomass retrieval with multi-source data: Prediction and applicability analysis in eastern Mongolia," *Land Degradation Develop.*, vol. 35, no. 9, pp. 2982–2992, May 2024, doi: 10.1002/ldr.5109.

[49] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973, doi: 10.1109/TSMC.1973.4309314.

[50] S. Pandit, S. Tsuyuki, and T. Dube, "Exploring the inclusion of Sentinel-2 MSI texture metrics in above-ground biomass estimation in the community forest of Nepal," *Geocarto Int.*, vol. 35, no. 16, pp. 1832–1849, Dec. 2020, doi: 10.1080/10106049.2019.1588390.

[51] *Calculation of Rangeland Carrying Capacity*, document NY/T 635-2015, Ministry of agriculture and rural affairs the People's Republic of China, Beijing, 2015.

[52] G. Nicolas, T. P. Robinson, G. R. W. Wint, G. Conchedda, G. Cinardi, and M. Gilbert, "Using random forest to improve the downscaling of global livestock census data," *PLoS ONE*, vol. 11, no. 3, Mar. 2016, Art. no. e0150424, doi: 10.1371/journal.pone.0150424.

[53] W. Wint and T. Robinson. (2007). *Gridded Livestock of the World 2007*. FAO. [Online]. Available: https://openknowledge.fao.org/handle/20.500.14283/A1259E

[54] D. Poobathy and R. M. Chezian, "Edge detection operators: Peak signal to noise ratio based comparison," *Int. J. Image, Graph. Signal Process.*, vol. 6, no. 10, pp. 55–61, Sep. 2014, doi: 10.5815/ijigsp.2014.10.07.

[55] S. Zhao, Y. Liu, R. Zhang, and B. Fu, "China's population spatialization based on three machine learning models," *J. Cleaner Prod.*, vol. 256, May 2020, Art. no. 120644, doi: 10.1016/j.jclepro.2020.120644.

[56] Y. Wang, C. Huang, M. Zhao, J. Hou, Y. Zhang, and J. Gu, "Mapping the population density in Mainland China using NPP/VIIRS and points-of-interest data based on a random forests model," *Remote Sens.*, vol. 12, no. 21, p. 3645, Nov. 2020, doi: 10.3390/rs12213645.

[57] N. Carvalhais et al., "Global covariation of carbon turnover times with climate in terrestrial ecosystems," *Nature*, vol. 514, no. 7521, pp. 213–217, Oct. 2014, doi: 10.1038/nature13731.

[58] T. Musavi et al., "Stand age and species richness dampen interannual variation of ecosystem-level photosynthetic capacity," *Nature Ecol. Evol.*, vol. 1, no. 2, p. 0048, Jan. 2017, doi: 10.1038/s41559-016-0048.

[59] B. Zhang, L. Zhang, D. Xie, X. Yin, C. Liu, and G. Liu, "Application of synthetic NDVI time series blended from Landsat and MODIS data for grassland biomass estimation," *Remote Sens.*, vol. 8, no. 1, p. 10, Dec. 2015, doi: 10.3390/rs8010010.

[60] S. Gong et al., "Response of the weathering carbon sink in terrestrial rocks to climate variables and ecological restoration in China," *Sci. Total Environ.*, vol. 750, Jan. 2021, Art. no. 141525, doi: 10.1016/j.scitotenv.2020.141525.

[61] Y. Yao et al., "Spatiotemporal pattern of gross primary productivity and its covariation with climate in China over the last thirty years," *Global Change Biol.*, vol. 24, no. 1, pp. 184–196, Jan. 2018, doi: 10.1111/gcb.13830.

[62] B. Kong, H. Yu, R. Du, and Q. Wang, "Quantitative estimation of biomass of Alpine grasslands using hyperspectral remote sensing," *Rangeland Ecol. Manage.*, vol. 72, no. 2, pp. 336–346, Mar. 2019, doi: 10.1016/j.rama.2018.10.005.

[63] H. Ren, B. Zhang, and X. Guo, "Estimation of litter mass in non-growing seasons in arid grasslands using MODIS satellite data," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 222–230, Jan. 2018, doi: 10.1080/22797254.2017.1418186.

[64] T. Dube and O. Mutanga, "Investigating the robustness of the new Landsat-8 operational land imager derived texture metrics in estimating plantation forest aboveground biomass in resource constrained areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 108, pp. 12–32, Oct. 2015, doi: 10.1016/j.isprsjprs.2015.06.002.

[65] D. Lu, Q. Chen, G. Wang, L. Liu, G. Li, and E. Moran, "A survey of remote sensing-based aboveground biomass estimation methods in forest ecosystems," *Int. J. Digit. Earth*, vol. 9, no. 1, pp. 63–105, Jan. 2016, doi: 10.1080/17538947.2014.990526.

[66] J. A. Gallardo-Cruz et al., "Predicting tropical dry forest successional attributes from space: Is the key hidden in image texture?" *PLoS ONE*, vol. 7, no. 2, Feb. 2012, Art. no. e30506, doi: 10.1371/journal.pone.0030506.

[67] R. W. Tuan, "Research on hyperspectral and multispectral image fusion," Ph.D. thesis, College Elect. Inf. Eng., Hunan Univ., Hunan, China, 2020, doi: 10.27135/d.cnki.ghudu.2020.002186.

[68] B. Hu, Q. Li, and A. Smith, "Noise reduction of hyperspectral data using singular spectral analysis," *Int. J. Remote Sens.*, vol. 30, no. 9, pp. 2277–2296, May 2009, doi: 10.1080/01431160802549344.

[69] Y. Kawakami and M. Sugiyama, "Investigating overparameterization for non-negative matrix factorization in collaborative filtering," in *Proc. 15th ACM Conf. Recommender Syst.* Amsterdam, The Netherlands: ACM, Sep. 2021, pp. 685–690, doi: 10.1145/3460231.3478854.

[70] A. Hore and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 2366–2369, doi: 10.1109/ICPR.2010.579.

[71] D. Kong, C. Ding, and H. Huang, "Robust nonnegative matrix factorization using L21-norm," in *Proc. 20th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2011, pp. 673–682, doi: 10.1145/2063576.2063676.

[72] S. Madonsela, M. Cho, A. Ramoelo, and O. Mutanga, "Investigating the relationship between tree species diversity and Landsat-8 spectral heterogeneity across multiple phenological stages," *Remote Sens.*, vol. 13, no. 13, p. 2467, Jun. 2021, doi: 10.3390/rs13132467.

[73] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911, doi: 10.1109/TGRS.2023.3336791.

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[74] X. Wu, "Analysis on the spatiotemporal changes of livestock in Qinghai grassland in recent 30 years," M.S. thesis, College Geograph. Sci., Qinghai Norm Univ., Qinghai, China, 2020, doi: 10.27778/d.cnki.gqhzy.2020.000517.

[75] T. Liang et al., "Multi-factor modeling of above-ground biomass in Alpine grassland: A case study in the three-river headwaters region, China," *Remote Sens. Environ.*, vol. 186, pp. 164–172, Dec. 2016, doi: 10.1016/j.rse.2016.08.014.

[76] H. Yu et al., "A method to avoid spatial overfitting in estimation of grassland above-ground biomass on the Tibetan Plateau," *Ecolog. Indicators*, vol. 125, Jun. 2021, Art. no. 107450, doi: 10.1016/j.ecolind.2021.107450.

[77] R. Teague and S. Dowhower, "Links of microbial and vegetation communities with soil physical and chemical factors for a broad range of management of tallgrass Prairie," *Ecolog. Indicators*, vol. 142, Sep. 2022, Art. no. 109280, doi: 10.1016/j.ecolind.2022.109280.

[78] J. Wang et al., "Effects of grazing exclusion on soil respiration components in an Alpine meadow on the north-eastern Qinghai-Tibet Plateau," *Catena*, vol. 194, Nov. 2020, Art. no. 104750, doi: 10.1016/j.catena.2020.104750.

[79] A. C. Risch, M. F. Jurgensen, and D. A. Frank, "Effects of grazing and soil micro-climate on decomposition rates in a spatio-temporally heterogeneous grassland," *Plant Soil*, vol. 298, nos. 1–2, pp. 191–201, Sep. 2007, doi: 10.1007/s11104-007-9354-x.

[80] E. Medina-Roldán, J. Paz-Ferreiro, and R. D. Bardgett, "Grazing exclusion affects soil and plant communities, but has no impact on soil carbon storage in an upland grassland," *Agricult., Ecosyst. Environ.*, vol. 149, pp. 118–123, Mar. 2012, doi: 10.1016/j.agee.2011.12.012.

[81] Z. Wang et al., "Quantitative assess the driving forces on the grassland degradation in the Qinghai–Tibet Plateau, in China," *Ecolog. Informat.*, vol. 33, pp. 32–44, May 2016, doi: 10.1016/j.ecoinf.2016.03.006.

[82] F. Dai, Z. Su, S. Liu, and G. Liu, "Temporal variation of soil organic matter content and potential determinants in Tibet, China," *CATENA*, vol. 85, no. 3, pp. 288–294, Jun. 2011, doi: 10.1016/j.catena.2011.01.015.

[83] S. R. Verón, J. M. Paruelo, and M. Oesterheld, "Grazing-induced losses of biodiversity affect the transpiration of an arid ecosystem," *Oecologia*, vol. 165, no. 2, pp. 501–510, Feb. 2011, doi: 10.1007/s00442-010-1780-4.

[84] H. Shen, S. Dong, A. DiTommaso, J. Xiao, and Y. Zhi, "N deposition may accelerate grassland degradation succession from grasses- and sedges-dominated into forbs-dominated in overgrazed Alpine grassland systems on Qinghai-Tibetan Plateau," *Ecolog. Indicators*, vol. 129, Oct. 2021, Art. no. 107898, doi: 10.1016/j.ecolind.2021.107898.

[85] R. T. Conant and K. Paustian, "Potential soil carbon sequestration in overgrazed grassland ecosystems," *Global Biogeochem. Cycles*, vol. 16, no. 4, pp. 90-1–90-9, Dec. 2002, doi: 10.1029/2001GB001661.

[86] H. Cai, X. Yang, and X. Xu, "Human-induced grassland degradation/restoration in the central Tibetan Plateau: The effects of ecological protection and restoration projects," *Ecolog. Eng.*, vol. 83, pp. 112–119, Oct. 2015, doi: 10.1016/j.ecoleng.2015.06.031.

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